

CEE 406 Pavement Design I
Department of Civil and Environmental Engineering
University of Illinois at Urbana-Champaign
Homework 10
Due: December 11th, 2024

1. Airfield Pavements

As an airport engineer,

- a. You were asked to check the runways of the airport for the operation of the Boeing B747-400. The tire pressure for the aircraft is 200 psi. The runways have PCR of 700/R/C/Y/T, 650/F/C/Y/T, and 600/F/B/X/T. On which runway(s) can the aircraft operate?

ACR for flexible pavement:

Subgrade Category	Subgrade Modulus [psi]	Flexible ACR Number	ACR Thickness t [in]
D	7,251.89	832.93	37.56
C	11,603.02	607.10	28.54
B	17,404.53	518.34	23.22
A	29,007.55	473.80	18.04

ACR for rigid pavement:

Subgrade Category	Subgrade Modulus [psi]	Rigid ACR Number	ACR Thickness t [in]
D	7,251.89	855.48	18.12
C	11,603.02	767.25	16.49
B	17,404.53	685.56	14.99
A	29,007.55	581.60	13.08

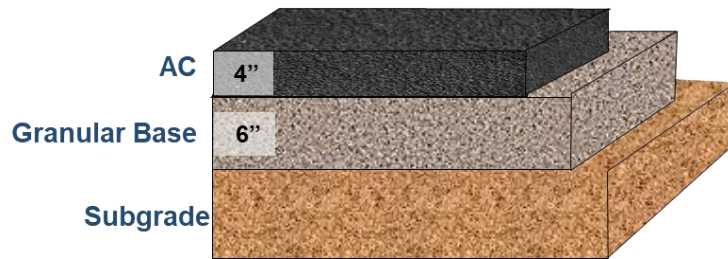
- b. Using FAARFIELD, please design the subbase layer thickness for a new flexible pavement to allow the operation of the Boeing B747-400. 4000 departures are forecasted annually. The subgrade CBR is 5%, and it can be increased to 7% by stabilization. The cost of stabilization is equivalent to the cost of 15in subbase. Please upload screenshots of your inputs and results.

2. Pavement Life Cycle Assessment

The traffic per lane on a 1-mi flexible pavement section (shown below) is **1000 vehicle/day** with no growth. The pavement deterioration is presumed to be linear over a **20-year analysis period**. The initial IRI of the pavement at construction is **60 in/mi**. It is assumed to increase linearly at a rate of **12.2 in/mi/year**. The agency decided to rehabilitate the pavement with a conventional mill and overlay every time the IRI reaches **170 in/mi**. When the rehabilitation is done, the IRI goes back to 60 in/mi that year. On average, each vehicle consumes **0.05 gallons of fuel for every mile of travel**.

Using the information presented in the table below, please conduct a complete pavement LCA for the **analysis period** to calculate GHG emissions for **1 lane-mile**. Please consider all pavement life cycle stages. Assume any missing information. **Comment** on the stage that contributes the most to GHG emissions and **describe** an approach to reduce the emission from that stage.

Pavement Section



Inventory

Item	GHG Emissions (kg CO ₂ -eq)	Assumptions/Notes
1 ton of AC mixture production	70	Density = 140 pcf
1 ton of aggregate production	8.1	Density = 130 pcf
Construction of 1 ton of AC	48	Density = 140 pcf
Construction of 1 ton of granular base	39	Density = 130 pcf
Transportation of 1 ton-mi of materials from plant to site	0.115	10 mi distance
1 gallon of fuel consumption for 1 vehicle	9	
Mill and overlay of 1 lane-mile of pavement	80,000	Number of rehabilitations depends on IRI
Disposal of 1 lane-mile of pavement	876	